Sets - 5 functions

Thursday, 14 March 2024 14:22

① a) IsMotherOf : not she can have more than kid

IsUncleOf : not a function

IsMarriedTo: a function a person can only be lawfully married to 1 person

b) IsMotherOf^{-1}: a function, a person has exactly 1 biological mother

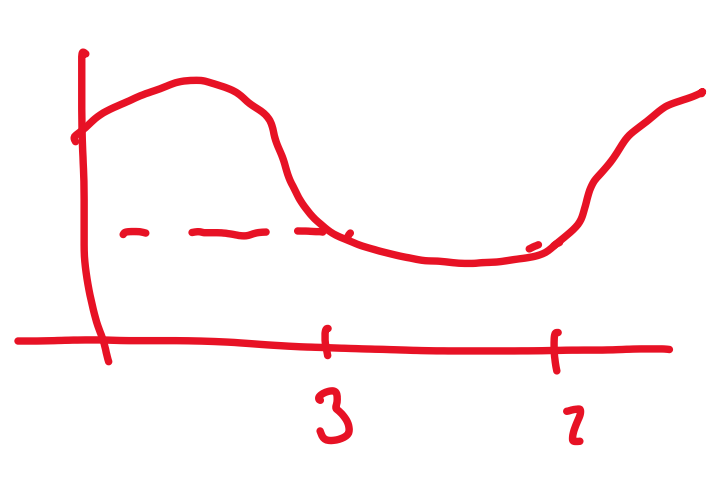
IsUncleof^{-1}: not a function a person can have multiple uncles

IsMarriedto^{-1}: the same relation as IsMarriedTo → a function

② f: A→ IN $A=\{10,11,12 ... 99\}$

f(n) $f(29)=2+9=11$

NOT INJECTIVE $f(29)=f(92)$



Horizontal LINE TEST

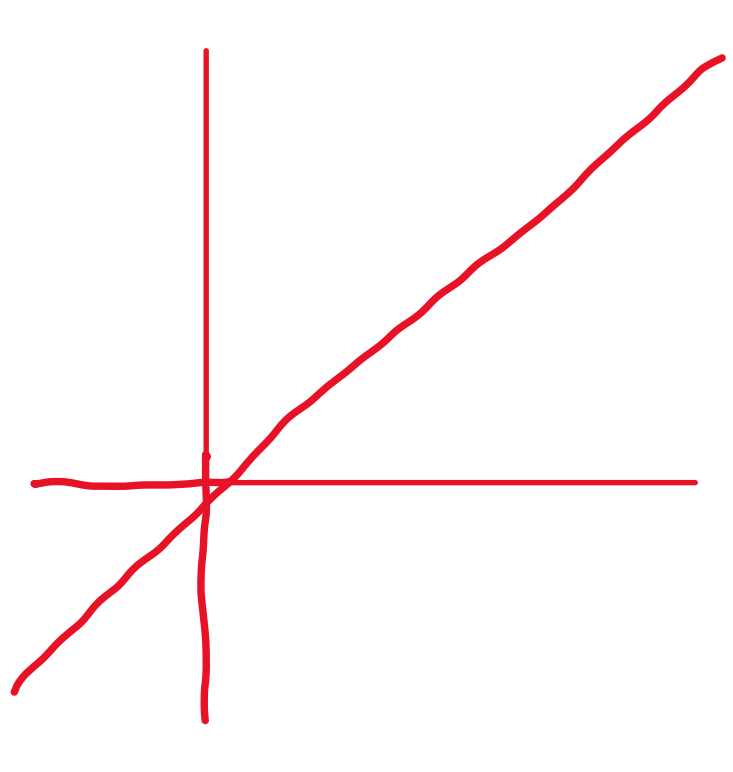
b) f: W→W f(this) = uijt

f(zero) = afsp

INJECTIVE : the same output (word) cannot come from two different input(words)

c) It's NOT INJECTIVE

③ a) f: IR→IR f(x)=x

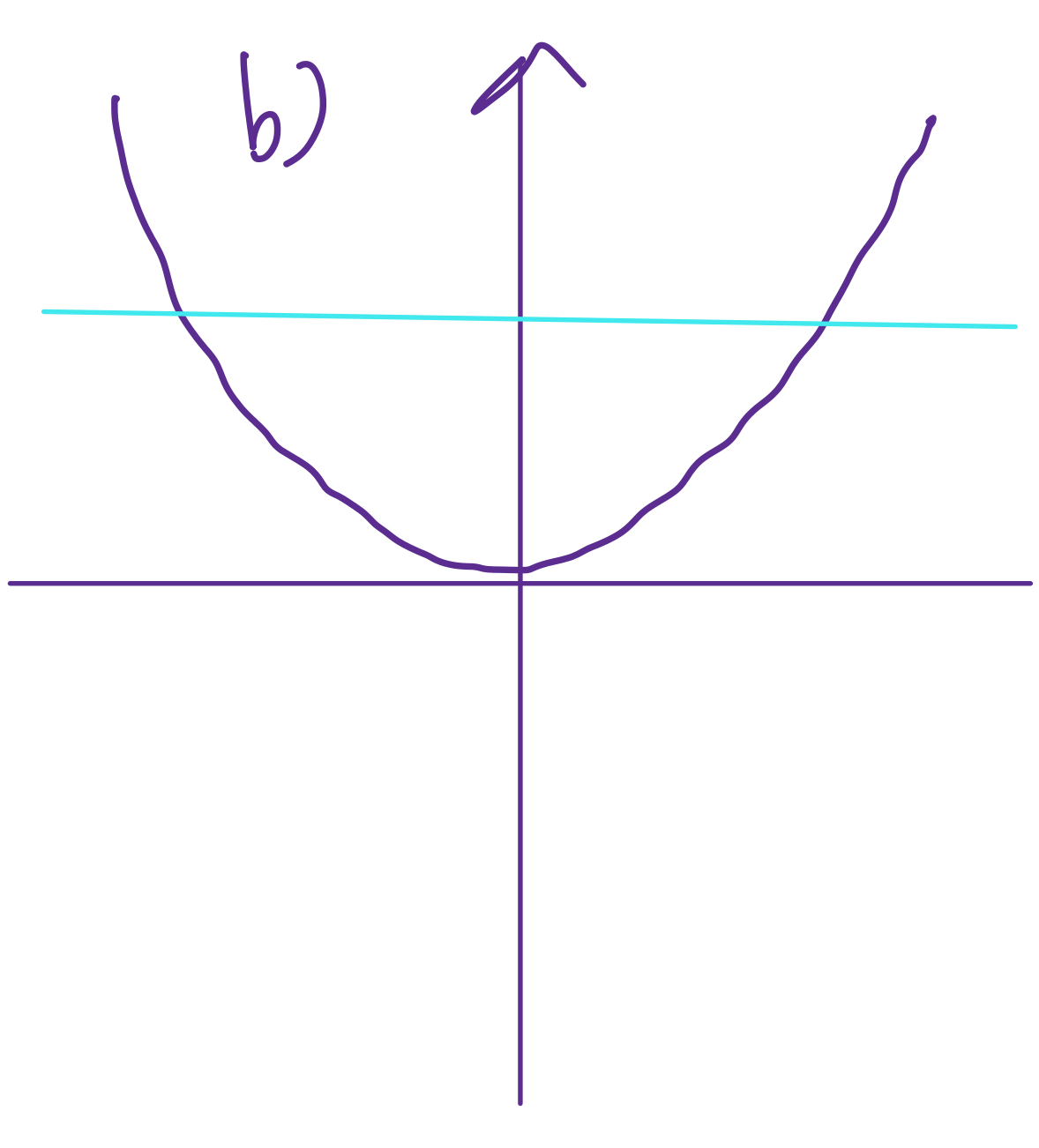


INJECTIVE $f(x)=f(y)$ $x=y$

SURJECTIVE ∀y ∈IR f(x)=y

If we take x=y all of IR is covered

b)



NOT INJECTIVE

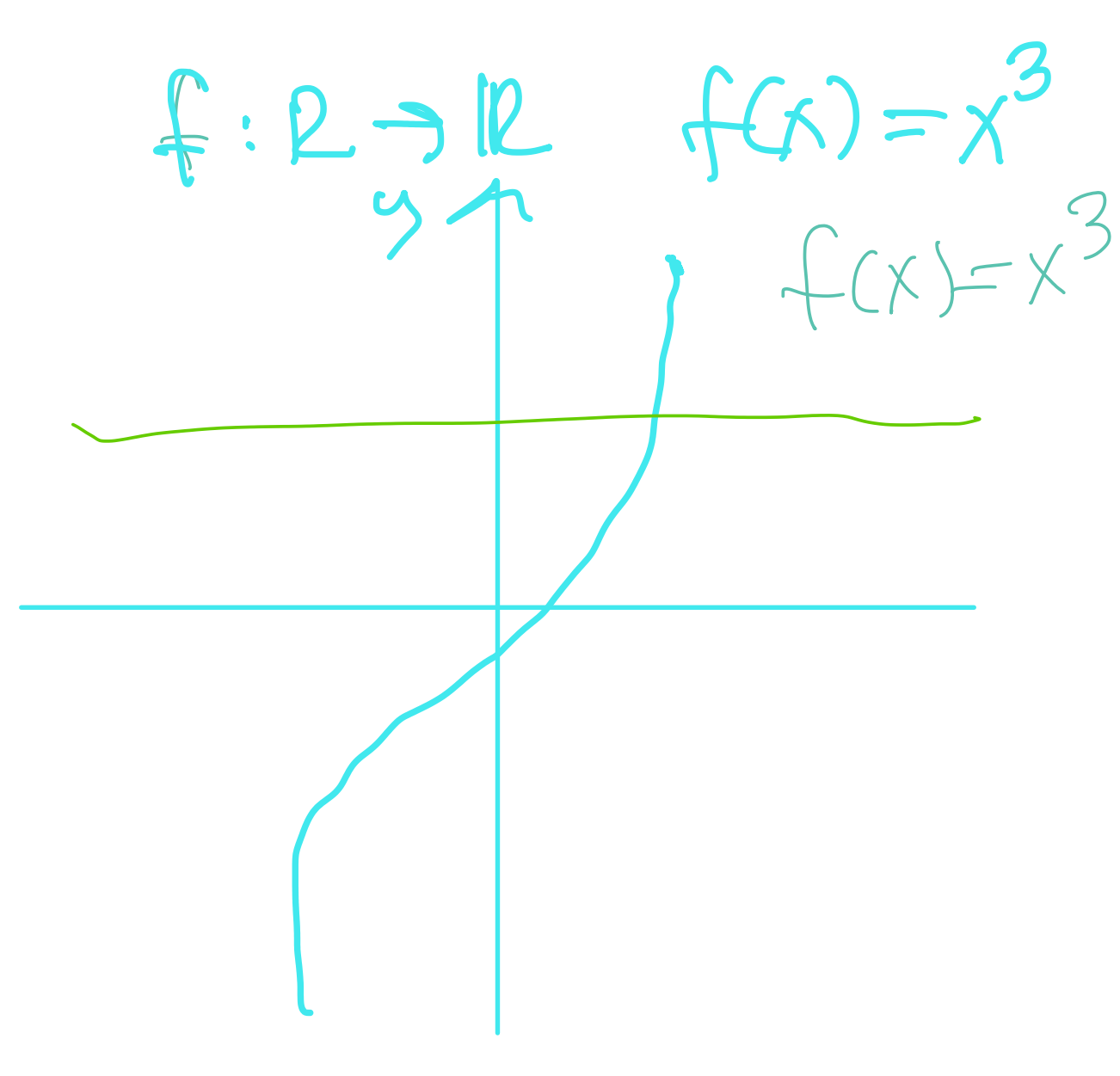
$f(-1)=f(1)=1$

NOT SURJECTIVE

$-1 \in IR_f$

$\Rightarrow R_f \neq IR$

c) f: R→IR $f(x)=x^3$



INJECTIVE :

$x<y \rightarrow f(x)=f(y)$

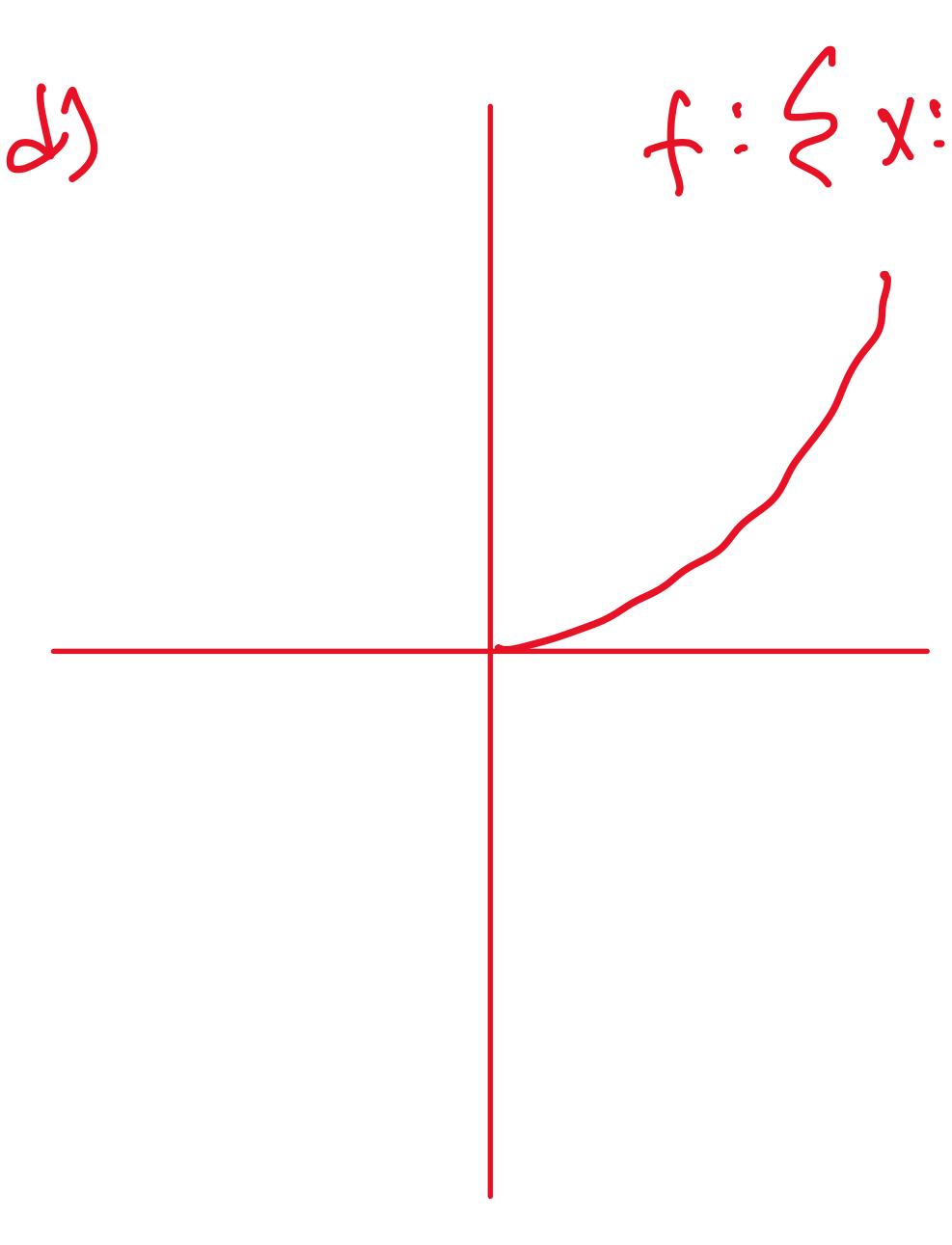
strictly increasing

SURJECTIVE :

$\forall y \in IR : f(x)=y$

If $x=y^{\frac{1}{3}}$

d) $f:\{x: x \geq 0\} \rightarrow IR$ $f(x)=x^2$



$f(x)=x^2$

INJECTIVE :

$x<y \rightarrow f(x)<f(y)$

strictly increasing

NOT SURJECTIVE :

$-1 \in R_f$

$\rightarrow R_f \neq IR$

④ a) f: IR→IR $f(x)=x+3 \rightarrow D_f=R_f=IR$

$y=x+3$

$x=y+3$

$\rightarrow y=x-3 \rightarrow f(x)^{-1}=x-3$

$D_{f^{-1}}=D_{f1}=IR$

b) f: IR→IR $f(x)=3x$ $\rightarrow D_f=R_f=IR$

$y=3x$

$x=3y$

$y=\frac{x}{3} \rightarrow f(x)^{-1}=\frac{x}{3}$

$D_{f-1}=R_{f-1}=IR$

c) $f:\{x: x \geq 0\}$ IR $f(x)=x^2$

$\rightarrow D_f=\{x: x \geq 0\}$

$R_f=\{y: y \geq 0\}$

$y=x^2$

$x=y^2$

$y=\sqrt{x}$ $f(x)^{-1}=\sqrt{x}$

$D_{f-1}=R_f=\{x: x \geq 0\}$

$R_{f-1}=D_f=\{y: y \geq 0\}$

d) $f:\{x: x \leq 0\} \rightarrow IR$ $f(x):=x^2$

$\rightarrow D_f=\{x: x \leq 0\}$

$R_f=\{x: x \geq 0\}$

$y=x^2$

$x=y^2$

$y=-\sqrt{x} \rightarrow f^{-1}(x)=-\sqrt{x}$

$D_{f-1}=R_f=\{x: x \geq 0\}$

$R_{f-1}=D_f=\{x: x \leq 0\}$

⑤ W_2

W_{2+}

$tr: W_{2+} \rightarrow W_2$

$index: W_2 \rightarrow IN$

$f: W_{2+} \rightarrow IN$

$f = index \circ tr$

tr(tree)=tr

index(aa)=1

index(af)=6

index(ba)=27

a) f takes as input a word of at least 2 letter, truncates that word to the first 2 letters and then returns the index of that word 2 letters lowercase

b) total: $D_f=A$ domain

Any word in W_{2+} can be truncated to a word in W_2 and every word in W_2 has an index

→f is total (it assigns an index to every word in W_{2+})

c) Subjective: $R_f=B$ codomain

NOT SUBJECTIVE.

$R_f = 26^2 = 676 \Rightarrow R_f \neq IN$

The index possibilities of a 2 letter word do not cover all of IN

⑥ a)

$f = sqrt \circ add_1 \circ mul_2 =$

$= sqrt(add_1(mul_2(x))) =$

$= sqrt(add_1(2x)) =$

$= sqrt(2x+1)$

$= \sqrt{2x+1}$

$D_f: 2x+1 \geq 0$

$x \geq -\frac{1}{2}$

$\rightarrow D_f=\{x: x \geq -\frac{1}{2}$

$D_f=\{y: y \geq 0\}$

$g = add_2^{-1} \circ sqrt^{-1} \circ mul_5 =$

$= add_2^{-1}(sqrt^{-1}(mul_5(x))) =$

$= add_2^{-1}(sqrt^{-1}(5x))$

$= add_2^{-1}((5x)^2) =$

$= (5x)^2 - 2$

$D_g = \{x: x \geq 0\}$

$R_g: y=(5 \cdot 0)^2-2=-2$

$R_g=\{y: y \geq -2\}$

b)

$f = mul_5^{-1} \circ add_2 \circ mul_3$

$f^{-1} = (mul_5^{-1} \circ add_2 \circ mul_3)^{-1} =$

$= mul_3^{-1} \circ add_2^{-1} \circ mul_5 = \frac{5x-2}{3}$

$\rightarrow D_{f^{-1}} = R_{f-1} = IR$

$g = sqrt^{-1} \circ add_1$

$g^{-1} = (sqrt^{-1} \circ add_1)^{-1} =$

$= add_1^{-1} \circ sqrt = \sqrt{x}-1$

$\rightarrow D_{g^{-1}} = \{x: x \geq 0\}$

$R_{g-1}: y=\sqrt{0}-1=-1$

$R_{g-1}=\{y: y \geq -1\}$

c) f: IR→IR $f(x)=2\sqrt{x-3} =$

$= 2\sqrt{add_3^{-1}(x)}$

$= 2(sqrt(add_3^{-1}(x)))$

$= mul_2(sqrt(add_3^{-1}(x)))$

$= mul_2 \circ sqrt \circ add_3^{-1}$

g: IR→IR $g(x)=(3\sqrt{x}+2)^2 =$

$= (3\, sqrt(x)+2)^2$

$= (mul_3(sqrt(x))+2)^2$

$= add_2(mul_3(sqrt(x)))^2 =$

$= sqrt^{-1}(add_2(mul_3(sqrt(x))))$

$= sqrt^{-1} \circ add_2 \circ mul_3 \circ sqrt$

7) $A:\{1,2. ...8\}$

$B=\{6,8,10...20\}$

f: A→B

$y=ax+b$

$6=a+b$

$8=2a+b$

$8=12-b$

$b=4$

$a=2$

$b=4$

$2x+4=f(x)$